Luke Zhuang

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Portfolio Linkedin Github

EDUCATION

California Polytechnic State University, San Luis Obispo
B.S. *Computer Science* to be conferred March 2025

COURSEWORK

- Software Engineering
- Data Structures
- Computer Organization
- Artificial Intelligence
- Systems Programming
- Object-Oriented Programming

SKILLS

- **Tools/Frameworks:** AWS, GitHub Actions, React.js, Node.js, MongoDB, SQL, Cloud Computing, Continuous Deployment
- **Programing:** C, Java, Python, JavaScript, HTML5, CSS3
- **Libraries/Technology:** Sass, JWT, Axios, Bcrypt, Figma
- **Operating Systems:** Linux, Windows, macOS, UNIX

PROJECTS

Inventory Management App: Live Site | Github
React, Node.js, TypeScript, Express.js, MongoDB, AWS, JWT, Axios, Bcrypt, Sass

- Implemented secure user authentication with password encryption (Bcrypt), reducing potential security vulnerabilities by 32%
- Deployed application assets and data on Amazon S3, improving data retrieval speed by an estimated 26%
- Integrated TypeScript to enforce type safety across the codebase, enhancing code readability and reducing runtime errors.

Blackjack: Live Site | Github
JavaScript, HTML5, CSS3

- Developed a fully responsive Blackjack game supporting desktop/mobile
- Implemented authentic gameplay features—splitting hands, doubling bets, and special Blackjack payouts—increasing user engagement

Server-and-Client: Github
UNIX, C

- Developed a lightweight web server in C supporting a targeted subset of HTTP methods for efficient request-response handling.
- Implemented child process forking to service concurrent client requests.

Matrix-Multiplication: Github
ARMv8-Assembly

- Parallelized matrix multiplication using SIMD instructions and multithreading, demonstrating a significant performance boost